Trustworthy Collective Reasoning and Evaluation for AI Agents

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1 The problem

Five AI agents read shipping reports and are asked whether a cargo vessel has left port. Three say yes and two say no, but the vessel is still in port. Two of the yes answers rest on the same outdated schedule, and the third comes from a compromised agent that argues confidently for yes. A majority vote picks the wrong answer, and open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source back each other up.

The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions.

The surest way is to verify each claim, by a person such as an analyst or a direct measurement such as a satellite image, but both are slow and costly, and AI agents produce claims far faster than people can verify them. AI judges can verify at scale, but a single judge has known biases, such as favoring longer answers, and can be misled by the answers it grades (Zheng et al., 2023). Debate helps the judge: two agents argue for opposite answers, exposing each other's flaws, until the dispute narrows to one step the judge can verify (Irving et al., 2018). Among AI agents, however, debate is open to the three failures in the example: agents that share a mistake seem to confirm each other, a dishonest agent can argue, rate others, or judge strategically, and persuasion can beat evidence.

2 Why it matters

AI models now grade one another's work, and multi-agent systems decide by vote or discussion (Zheng et al., 2023; Verga et al., 2024). All three failures occur. Models' errors are correlated, more so as models improve (Kim et al., 2025; Goel et al., 2025), so agreement is weaker evidence than it looks: in one study, nine AI judges were worth about two independent ones (Kohli, 2026). Agents collude: pairs verifying each other's work colluded in 94% of runs when breaking the rules paid more (Shi et al., 2026). Discussion erases the disagreement that exposes errors: a single round of discussion made a panel nearly unanimous while barely changing its accuracy (Shu, 2026).

Navy decisions face all of these problems at once: AI tools differ in reliability that is rarely known in advance, the true answer often arrives late, an adversary may compromise an agent or plant false reports, and verification is costly or impossible when connectivity is limited. A commander must still decide whether to act on what the agents agree on.

3 How it is solved today, and the limitations

Statistical methods estimate each rater's accuracy without an answer key (Dawid and Skene, 1979; Whitehill et al., 2009) and detect sources that copy one another (Dong et al., 2009), but they assume that raters err independently or in simple patterns (Jaffe et al., 2016) and never change their minds. Recent work estimates the shared mistakes of AI judges but treats them as honest and as never conferring (Hossain et al., 2026; Zhu et al., 2026).

Computational social choice designs impartial peer review, in which a participant's ratings cannot

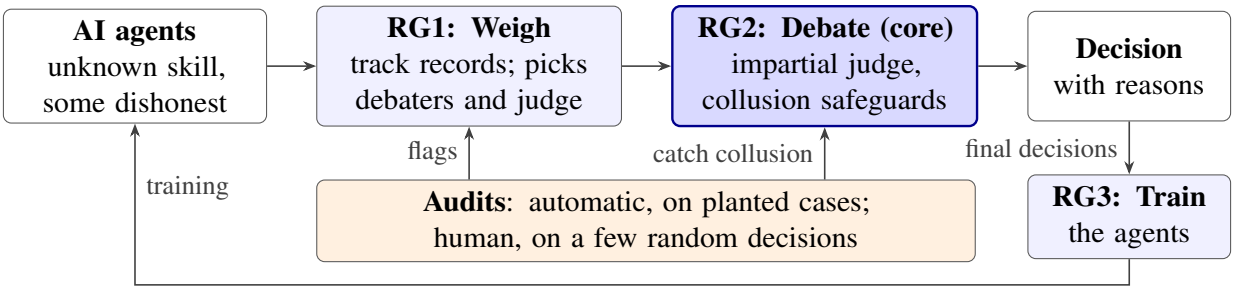


Figure 1: RG1 weighs answers and picks debaters and a judge; RG2, the core, replaces open discussion with a debate harness. Automatic and human audits catch collusion; RG3 trains agents on final decisions.

change its own chance of selection (Kurokawa et al., 2015; Alon et al., 2011), and fair panel selection (Flanigan et al., 2021). Lev et al. (2023) added reliability weights to impartial selection, but their weights reward agreeing with the majority, which fails once accurate reviewers are no longer the majority. These mechanisms take opinions as given; in our setting some claims can be verified, and opinions change as agents talk.

Multi-agent debate can help (Du et al., 2024), but majority voting explains most of the gain (Choi et al., 2025), and one persuasive adversary can sharply cut a group's accuracy (Amayuelas et al., 2024). A recent collaborative alternative helps weak judges more than standard debate does (Irving et al., 2018; Khan et al., 2024), but its judges accept the models' consensus even when both are wrong, and it assumes honest models (Jiang et al., 2026). No existing debate protocol holds up when agents share mistakes, lie, or collude.

4 Our approach: what is new, and why it will succeed

Our approach centers on debate and guards it against all three failures (Figure 1). Each agent first answers alone, with its evidence. Research goal 1 (RG1) weighs the answers by each agent's track record and picks debaters and judges impartially. If the weighted answers still disagree, the group does not discuss until it agrees: two representatives debate before a judge, and the judge's choice is the group's decision (RG2). Making this debate trustworthy is the core of the project: the judge has no stake in either side, any tool evidence must be re-runnable, and automatic audits (planted cases with known answers) and human audits (a few random decisions) catch debaters or judges who collude. RG3 trains agents on the group's final decisions, which needs no human labels.

RG1. Decide what to debate, and pick debaters and judges impartially

A plain vote lets shared mistakes pile up, and agents choosing debaters can favor themselves.

Task 1.1: Deciding what to debate, counting shared mistakes once. Each answer counts according to its agent's track record, estimated from who agrees with whom across real questions, as one can grade an exam without the answer key; agents flagged by audits (Task 2.2) count less. The core method extends the rater model of Dawid and Skene (1979) with shared error factors: agents that keep giving the same unusual answers share a factor, and their agreement adds little evidence. In the vessel example, the two agents sharing an outdated schedule count about once, so yes no longer leads; with no clear leader, the question goes to debate.

Task 1.2: Impartial choice of debaters and judges. A debate needs a judge and one representative per answer. Representatives are chosen by peer ratings of their arguments, with raters weighted

by track record, as in our PeerEval leaderboard (peereval-ai.github.io) and PRD method (Li et al., 2024a). So that no agent can rate its way into a role, we adapt the partition idea of impartial peer review (Alon et al., 2011; Kurokawa et al., 2015): agents sharing an underlying model form a group, rated only by outsiders. Each answer's representative is the top-rated member of a randomly drawn group that gave that answer; the judge is a random agent that did not answer and shares no model with either side. Ratings pick only who argues; the judge picks the answer.

RG2. Settle disputes by agentic debate, and catch collusion

Open discussion erases disagreement and rewards persuasion, and debaters can collude.

Task 2.1: Debate before an impartial judge. Today's multi-agent systems discuss until they agree, which erases the disagreement that exposes errors (Shu, 2026). We instead run debate in a harness, the software around models that turns them into agents: a sandbox with tools, a control loop that decides what happens next, memory, and verification of results (Ning et al., 2026). Our control loop sets who speaks, who judges, and when to stop: Task 1.2 picks one representative per answer and a judge with no stake in either, the two walk through their arguments to the first step on which they differ (Irving et al., 2018), and the loop stops there for the judge. The judge's choice is the group's decision; most LLM debates instead run fixed rounds of text between given models. Tools are optional, but tool evidence must carry its exact call, which the judge, itself an agent (Zhuge et al., 2025), re-runs in the sandbox, whereas current tool-using debates only weigh arguments (Jeong et al., 2026). This builds on PRD's peer discussion and HiPRAG's search agents, and the harness logs each debate for audits and training. In the vessel example, the no side's query returns a newer port report listing the vessel at its berth; the judge re-runs it and picks no.

Task 2.2: Automatic and human audits against collusion. Collusion here means debaters, or the judge, steering a debate to the wrong side. A debater can argue its side badly on purpose, or colluders can mislead the judge with true but selectively chosen facts (Hu et al., 2026), which re-running evidence does not catch. Automatic audits mix in planted debates on planted cases, kept out of training, such as a paper with a deliberately inserted error, as in studies of human peer review (Schroter et al., 2008). Debaters and judges whose planted debates keep ending on the wrong side are flagged. Because capable models can often tell evaluations from real use (Needham et al., 2025), a colluder could behave well only in planted debates. Human audits therefore cover a few real decisions chosen at random: a person (such as an analyst) or a direct measurement (such as a satellite image) finds the right answer. As with random tax audits, no agent knows which decisions will be audited, and agents that do worse on audited decisions than on planted ones are flagged.

RG3. Train agents that contribute evidence

Rewarding persuasion has produced agents that fabricate quotations (Kenton et al., 2026) and mislead people (Wen et al., 2025).

Task 3: Rewards for verified contributions. We will train agents with reinforcement learning on the group's final decisions, rewarding each message for moving the group toward the decision; tool-backed decisions weigh more, and decisions involving flagged agents are dropped. This extends our process reward models, FG-PRM and HiPRAG (Li et al., 2025; Wu et al., 2026), from one agent's steps to many agents' messages. The main experiment compares this with the usual reward for winning over the judge (Arnesen et al., 2024).

Evaluation: testbeds and a red team

All data are synthetic or public. We will use (A) math and code problems solved by agents of known skill, some sharing a flawed hint; (B) multi-document questions from our MEQA pipeline (Li et al., 2024b), including synthetic maritime scenarios; and (C) scientific claims and peer review, with reproducibility studies as audits. In the testbeds, a human audit reveals the stored answer, within a fixed budget per method. Following AI control (Greenblatt et al., 2024), a red team will train debaters and judges to collude, to spot planted debates, and to cheat only when they expect no audit, and we will measure how many wrong decisions get through at each audit budget. Baselines are majority vote, our PRD, text-only and tool-using debate (Jiang et al., 2026; Jeong et al., 2026), and judge-weighting methods (Hossain et al., 2026; Zhu et al., 2026; Lev et al., 2023).

What is new. (i) In place of open discussion, a debate harness that stops at the disputed step for an impartial judge, who can re-run tool evidence. (ii) Automatic audits on planted debates and a few human audits that expose colluding debaters or judges, even those behaving well only in planted debates. (iii) Weights that count shared mistakes about once. (iv) Training on the group's own decisions, weighted by evidence, without human labels.

Why it will succeed. Theory supports the design: in debate, a judge who can verify one disputed step can settle arguments far beyond its own ability (Irving et al., 2018), and our judge, who has no stake in either side, rules on that step. Without trusted audits, dishonest agents that imitate honest ones cannot be told apart from them, but a number of audits that does not grow with the number of items suffices, even when most agents are dishonest (Steinhardt et al., 2016). Most tasks build on the PI's own tools. For example, PRD agrees better with human judgments than the strongest single AI judge (Li et al., 2024a), and HiPRAG cuts unnecessary searches to 2.3% (Wu et al., 2026). The PI built PeerEval, the first leaderboard on which AI models rank one another, co-organized the AI4Research workshop at AAAI 2025, and co-wrote a survey of large language models for scientific research, including peer review (Luo et al., 2025). The PI also co-led one of five red teams in the Amazon Nova AI Challenge, holds a 2024 NSF CAREER award and a 2023 Amazon Research Award, and as a postdoctoral researcher led system integration for RESIN-11 in DARPA KAIROS.

5 Limitations of our approach

Where no tool applies, the judge rules on argument alone, as in standard debate. If every agent shares the same misconception, there is nothing to debate, and audits catch the error only by chance, while training would reinforce it. A group that cheats only once, on a single high-stakes decision, may succeed before it is caught; we can bound such losses but not rule them out. Agents may also coordinate through hidden signals (Mathew et al., 2025), so robustness is shown only for the attacks our red team tries. Navy use would need Navy data and expert validation.

6 Impact if successful

The project will show when a debate among AI agents can be trusted and how few audits it takes to keep the agents honest. A Navy analyst using several AI tools would get a decision with its reasons: whose judgments carried weight and why, which step decided the debate, and which evidence the judge reproduced. These methods would let conferences and funders try impartial, audited peer review by AI. We will release code, testbeds, and protocols.

References

N. Alon, F. Fischer, A. D. Procaccia, and M. Tennenholtz (2011). Sum of Us: Strategyproof Selection from the Selectors. TARK, pp. 101–110.

A. Amayuelas, X. Yang, A. Antoniades, W. Hua, L. Pan, and W. Y. Wang (2024). MultiAgent Collaboration Attack: Investigating Adversarial Attacks in Large Language Model Collaborations via Debate. Findings of EMNLP, pp. 6929–6948.

S. Arnesen, D. Rein, and J. Michael (2024). Training Language Models to Win Debates with Self-Play Improves Judge Accuracy. arXiv:2409.16636.

H. K. Choi, X. Zhu, and S. Li (2025). Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models? NeurIPS.

A. P. Dawid and A. M. Skene (1979). Maximum Likelihood Estimation of Observer Error-Rates Using the EM Algorithm. Journal of the Royal Statistical Society, Series C (Applied Statistics) 28(1):20–28.

X. L. Dong, L. Berti-Equille, and D. Srivastava (2009). Integrating Conflicting Data: The Role of Source Dependence. PVLDB 2(1):550–561.

Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch (2024). Improving Factuality and Reasoning in Language Models through Multiagent Debate. ICML, PMLR 235:11733–11763.

B. Flanigan, P. Gölz, A. Gupta, B. Hennig, and A. D. Procaccia (2021). Fair Algorithms for Selecting Citizens' Assemblies. Nature 596:548–552.

S. Goel, J. Strüber, I. A. Auzina, K. K. Chandra, P. Kumaraguru, D. Kiela, A. Prabhu, M. Bethge, and J. Geiping (2025). Great Models Think Alike and this Undermines AI Oversight. ICML, PMLR 267:19621–19678.

R. Greenblatt, B. Shlegeris, K. Sachan, and F. Roger (2024). AI Control: Improving Safety Despite Intentional Subversion. ICML, PMLR 235:16295–16336.

E. Hossain, N. Yousefi, and S.-N. Lim (2026). Agreement Overstates Evidence: Error Dependence in LLM Judge Consensus. arXiv:2609.22512.

J. Hu, X. Huang, Y. Sun, Y. Dong, and X. Huang (2026). Lying with Truths: Open-Channel Multi-Agent Collusion for Belief Manipulation via Generative Montage. ACL.

G. Irving, P. Christiano, and D. Amodei (2018). AI Safety via Debate. arXiv:1805.00899.

A. Jaffe, E. Fetaya, B. Nadler, T. Jiang, and Y. Kluger (2016). Unsupervised Ensemble Learning with Dependent Classifiers. AISTATS, PMLR 51:351–360.

S. Jeong, Y. Choi, J. Kim, and B. Jang (2026). Tool-MAD: A Multi-Agent Debate Framework for Fact Verification with Diverse Tool Augmentation and Adaptive Retrieval. arXiv:2601.04742.

Y. Jiang, C. Chen, T. Wu, L. Sun, H. Liu, S. Feng, and C. Tan (2026). Collaborative Disagreement Resolution for Scalable Oversight. ICML.

Z. Kenton, L. Janzer, R. Greig, T. H. Teh, K. Tyshchuk, J. Brown-Cohen, H. Edwards, S. Rajamanoharan, N. Y. Siegel, N. Jaques, and R. Shah (2026). Debate Training Reduces Reward Hacking in RLAIF. arXiv:2608.17776.

A. Khan et al. (2024). Debating with More Persuasive LLMs Leads to More Truthful Answers. ICML, PMLR 235:23662–23733.

E. M. Kim, A. Garg, K. Peng, and N. Garg (2025). Correlated Errors in Large Language Models. ICML, PMLR 267:30038–30066.

G. Kohli (2026). Nine Judges, Two Effective Votes: Correlated Errors Undermine LLM Evaluation Panels. arXiv:2605.29800.

D. Kurokawa, O. Lev, J. Morgenstern, and A. D. Procaccia (2015). Impartial Peer Review. IJCAI, pp. 582–588.

O. Lev, N. Mattei, P. Turrini, and S. Zhydkov (2023). PeerNomination: A Novel Peer Selection Algorithm to Handle Strategic and Noisy Assessments. Artificial Intelligence 316:103843.

R. Li, T. Patel, and X. Du (2024a). PRD: Peer Rank and Discussion Improve Large Language Model based Evaluations. TMLR.

R. Li, Z. Wang, S. Q. Tran, L. Xia, and X. Du (2024b). MEQA: A Benchmark for Multi-hop Event-centric Question Answering with Explanations. NeurIPS 37, Datasets and Benchmarks, pp. 126835–126862.

R. Li, Z. Luo, and X. Du (2025). FG-PRM: Fine-grained Hallucination Detection and Mitigation in Language Model Mathematical Reasoning. Findings of EMNLP, pp. 4247–4278.

Z. Luo, Z. Yang, Z. Xu, W. Yang, and X. Du (2025). LLM4SR: A Survey on Large Language Models for Scientific Research. arXiv:2501.04306.

Y. Mathew, O. Matthews, R. McCarthy, J. Velja, C. Schroeder de Witt, D. Cope, and N. Schoots (2025). Hidden in Plain Text: Emergence and Mitigation of Steganographic Collusion in LLMs. IJCNLP-AACL, pp. 585–624.

J. Needham, G. Edkins, G. Pimpale, H. Bartsch, and M. Hobbhahn (2025). Large Language Models Often Know When They Are Being Evaluated. arXiv:2505.23836.

X. Ning, K. Tieu, D. Fu, T. Wei, Z. Li, et al. (2026). Code as Agent Harness: Toward Executable, Verifiable, and Stateful Agent Systems. arXiv:2605.18747.

S. Schroter, N. Black, S. Evans, F. Godlee, L. Osorio, and R. Smith (2008). What Errors Do Peer Reviewers Detect, and Does Training Improve Their Ability to Detect Them? Journal of the Royal Society of Medicine 101(10):507–514.

X. Shi, Y. Zhang, and D. Yang (2026). Emergent Collusion in Long-Horizon LLM Agent Interaction. arXiv:2609.24967.

Y. Shu (2026). Unanimity Without Persuasion: A Single Round of Debate Erases the Disagreement That Verification Needs. arXiv:2609.26145.

J. Steinhardt, G. Valiant, and M. Charikar (2016). Avoiding Imposters and Delinquents: Adversarial Crowdsourcing and Peer Prediction. NeurIPS 29.

P. Verga et al. (2024). Replacing Judges with Juries: Evaluating LLM Generations with a Panel of Diverse Models. arXiv:2404.18796.

J. Wen, R. Zhong, A. Khan, E. Perez, J. Steinhardt, M. Huang, S. R. Bowman, H. He, and S. Feng (2025). Language Models Learn to Mislead Humans via RLHF. ICLR.

J. Whitehill, P. Ruvolo, T. Wu, J. Bergsma, and J. Movellan (2009). Whose Vote Should Count More: Optimal Integration of Labels from Labelers of Unknown Expertise. NeurIPS 22, pp. 2035–2043.

P. Wu, M. Zhang, K. Wan, W. Zhao, K. He, X. Du, and Z. Chen (2026). HiPRAG: Hierarchical Process Rewards for Efficient Agentic Retrieval Augmented Generation. ICLR.

L. Zheng et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS 36, Datasets and Benchmarks, pp. 46595–46623.

B. Zhu, Y. Xie, and Y. Rao (2026). Stopping and Routing LLM Judge Panels. arXiv:2608.19802.

M. Zhuge, C. Zhao, D. Ashley, W. Wang, D. Khizbullin, Y. Xiong, Z. Liu, E. Chang, R. Krishnamoorthi, Y. Tian, Y. Shi, V. Chandra, and J. Schmidhuber (2025). Agent-as-a-Judge: Evaluate Agents with Agents. ICML.